

3.1

10. What is the maximum number of vertices (internal and leaves) in an m -ary tree of height h ?

Recall by Theorem 3, $n = mi + l$. We know by Theorem 4, $l \leq m^h$. So, the maximum # of leaves is m^h . By the Corollary in 3.1, $i = \frac{l-1}{m-1}$. So,

$$\begin{aligned} n &= mi + l \\ &= \boxed{m \left(\frac{m^h - 1}{m - 1} \right) + m^h} \end{aligned}$$

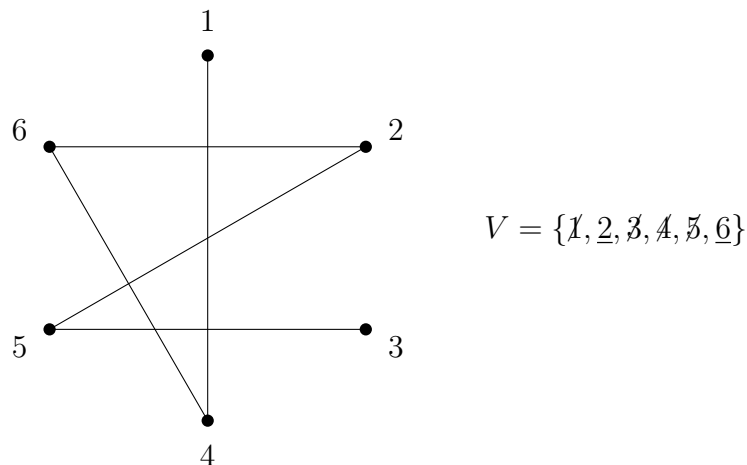
16. A *forest* is an unconnected graph that is a disjoint union of trees. If G is an n -vertex forest of t trees, how many edges does it have?

For any tree, it has $n - 1$ edges where n is the # of vertices in the tree. Since there are t trees in G , trees 1 will have $n_1 - 1$ edges, 2 with $n_2 - 1$ edges, and t with $n_t - 1$ edges where each n_i adds up to n . So, the # of edges in G should be

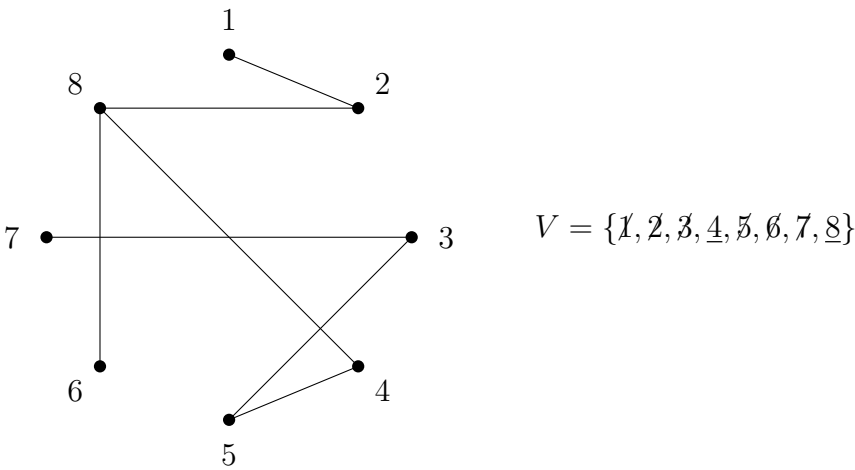
$$\begin{aligned} &\Rightarrow (n_1 - 1) + (n_2 - 1) + \dots + (n_t - 1) \\ &\Rightarrow \underbrace{(n_1 + n_2 + \dots + n_t)}_n - \underbrace{(1 + 1 + \dots + 1)}_{t \text{ times}} \\ &\Rightarrow \boxed{n - t} \end{aligned}$$

31. In the proof of Theorem 5, we showed that a Prufer sequence $(s_1, s_2, \dots, s_{n-2})$ uniquely described a tree on n items. Construct the trees with the following Prufer sequences.

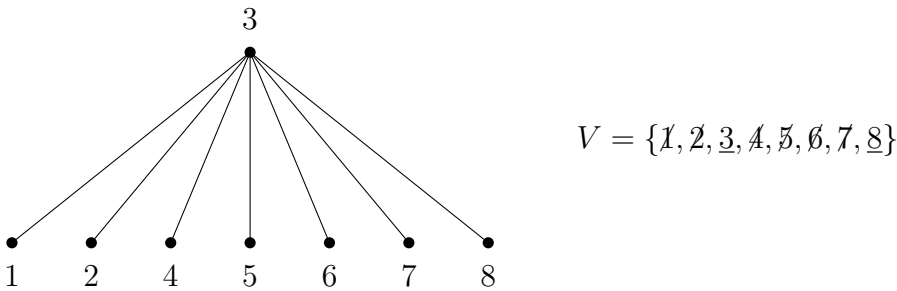
(a) (4, 5, 6, 2)



(b) (2, 8, 8, 3, 5, 4)



(c) (3, 3, 3, 3, 3, 3)

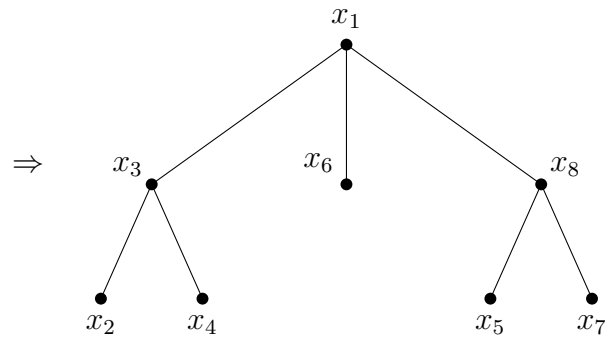


3.2

4. Test the graph whose adjacency matrix is given below to see if it is connected.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
x_1	0	0	1	0	0	1	0	1
x_2	0	0	1	0	1	0	1	0
x_3	1	1	0	1	0	0	0	0
x_4	0	0	1	0	1	1	1	0
x_5	0	1	0	1	0	0	0	1
x_6	1	0	0	1	0	0	0	0
x_7	0	1	0	1	0	0	0	1
x_8	1	0	0	0	1	0	1	0

* To see if a graph is connected, apply BFS



Since every vertex is visited, this graph is connected.